

Miguel Grisales Coca

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EDUCATION

University of Central Florida — Bachelor of Science in Mechanical Engineering | GPA: 3.18 August 2026
Intelligent Robotic Systems — Minor

EMPLOYMENT HISTORY

Assistive Robotics Lab — Dr. Aman Behal Sep. 2025 — Mar. 2026

Undergraduate Research Assistant,

- **Design and Prototyping:** Design and fabrication of components for Kinova Gen-3 robotic manipulator arm.

EMBRACE Lab — Dr. Ladda Thiamwong Oct. 2022 — Feb. 2025

Undergraduate Research Assistant, University of Central Florida, College of Nursing

- **Experimental Design:** Selected data collection strategies, wrote standard operating procedures, and translated instructions and tests for Spanish-speaking participants.
- **Participant Recruitment and Screening:** Recruited and screened older adults for fall risk study.
- **Data Collection:** Administered strength, mobility, body composition, and cognitive function assessments.

WEAR Lab — Dr. Joon-Hyuk Park Nov. 2021 — Apr. 2024

Undergraduate Research Assistant, University of Central Florida, College of Engineering and Computer Science

- **Experimental Design:** Created methodology to evaluate a lab-developed wrist mounted actigraphy device against commercially available devices. In partnership with EMBRACE Lab.
- **Design and Prototyping:** Lead designer on multiple projects using CAD, FEA, 3D printing, and machining.
- **Controls:** Developed algorithms for embedded controllers in C/C++.

SENIOR PROJECT

Bring ROAR to Life — Capstone Project “Hey, Anky!” July 2026

Physical-AI controlled 14-DOF quadrupedal dinosaur robot with reinforcement learning training and voice teleoperation.

Role — Controls Lead: Developed a Machine Learning policy and ROS2 nodes to control the robot.

- Deployed the team's CAD assembly in Nvidia's **Isaac Lab 3.0** to train a RL-policy to walk over millions of simulations.
- Created an inference method for the RL policy in a **ROS2** node running on a Raspberry Pi and **Linux Ubuntu**.
- Designed and implemented ROS2 nodes for reading and filtering sensor data (IMU, positional feedback, audio).
- Applied **MATLAB**'s *Robot Toolbox* to design a ROS2 compatible control backup using an inverse kinematics approach.

Role — Electrical Lead: Selected electronic components and designed power systems.

- Created a test to measure response time and estimate transfer function of servo motors assuming a second order system.
- Fabricated and tested protoboards which miniaturized the footprint of sensors and controllers.
- Designed power delivery and power safety schema for all systems.

Role — Technical Lead: Ensuring that the team reached its stated goals and providing mentorship.

- Mentored teammates and classmates regarding programming, controls, and electromechanical design.
- Led weekly meetings and discussions for longitudinal and time-sensitive project milestones.
- Facilitated task delegation and milestone reporting to monitor progress on individual and team levels.

SKILLS

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|----------------------------|-------------------------|------------------------------|-----------------------|------------------|
| • Digital Tools: | Solidworks (Proficient) | Isaac Lab (Intermediate) | MATLAB (Intermediate) | ANSYS (Familiar) |
| • Programming: | C/C++ (Intermediate) | Python (Basic) | ROS2 (Intermediate) | |
| • Embedded Systems: | Ubuntu (Proficient) | Power Systems (intermediate) | | |

AWARDS

- Faculty's Choice Award, University of Central Florida Senior Design Showcase (*Bring ROAR to Life*). July 2026
- T-LEARN Undergraduate Research Fellowship, University of Central Florida \$2,000. August 2022

PUBLICATIONS

- Choudhury, R., Park, J.-H., Banarjee, C., **Grisales Coca, M.**, Fukuda, D., Xie, R., Stout, J., and Thiamwong, L., 2024, “Associations between Monitor-Independent Movement Summary (MIMS) and fall risk appraisal combining fear

POSTERS

- Choudhury, R., Fournelis, N., **Grisales Coca, M.**, Dela Virgen, M. A., and Park, J.-H., 2023, “Pre-Impact Fall Detection and Balance Recovery Assistance Using Lower Limb Hybrid FES-Exoskeleton,” Department of Mechanical and Aerospace Engineering, University of Central Florida.
- Kadhim, I. F., **Grisales Coca, M.**, Sivakumaar, B., and Park, J.-H., 2023, “Mechatronic Musical Instrument (MMI),” Department of Mechanical and Aerospace Engineering, University of Central Florida.

ADDITIONAL PROJECTS

Airfoil Inspired Car — Eml 3303C Mechanical Engineering Measurements, University of Central Florida **July 2025**

Miniature car with airfoil inspired design for drag coefficient and flow analysis in a laser vapor wind tunnel.

Role – CAD Designer

- Analyzed airfoil (SD7062) drag polars to determine the best angle of attack to maximize downthrust -drag coefficient.
- Applied solidworks to model a car with an airfoil-shaped body for 3D printing.

Dynamic Swept Wing — WEAR Lab, University of Central Florida **April 2024**

Morphing airfoil for submersed flow analysis. User controlled telescopic span and variable sweep angle.

Role – Systems Integrator: Responsible for coalescing various pieces and prototypes into one deliverable.

- Designed power transmission, vice-locking cable tension mechanism, and controller housing in SolidWorks.
- Miniaturized electrical components and controller by modification of components and soldering.
- Optimized controls in C/C++ for speed, accuracy, and ease of use.

Soft Bodied Red-Seeking Worm — EGN 4060C Intro to Robotics, University of Central Florida **December 2023**

Soft bodied robot designed to emulate worm inching and color seeking behavior.

Role – CAD and Electronics Lead

- Designed the soft body’s geometry to promote left-right balance by increasing directional stiffness during contraction.
- Developed the power system for the actuators and controller.

Cable-driven Ankle Perturbation System — WEAR Lab, University of Central Florida **July 2023**

Cable driven system to measure the human gait response to trips and slips by suddenly and safely pulling on the ankle.

Role – CAD Designer

- Created 80/20 frame in SolidWorks and used FEA to prove a factor of safety of two with minimal deflection.
- Routed the cable system to pull an ankle in any planar direction.

Human Augmentation via Upper Lower-body Coupling — WEAR Lab, University of Central Florida **March 2023**

An unpowered exosuit to improve the ability of a person to jump by coupling the arms and legs mechanically.

Role – CAD Designer

- Modeled a prototype planetary gearbox with reverse engineered pull start system, and exosuit in SolidWorks.

Multimodal Human and Ambient Sensing System — WEAR Lab, University of Central Florida **September 2022**

Open-source and wrist mounted health and activity monitor with IMU and PPG functionality.

Role – Research Assistant

- Implemented a protocol to assess this device against commercially available actigraphy devices.
- Fabricated the housing and wrist mounting for electronics, minimizing overall footprint.

Mechatronic Musical Instrument — WEAR Lab, University of Central Florida **February 2022**

Cello designed for minimum size, assembly and disassembly, ease of use, and silence.

Role – CAD Designer

- Designed in SolidWorks and constructed with 3D printed parts, wood, and metal reinforcement.

LANGUAGES

English	Speaking: Advanced	Writing: Advanced	Reading: Advanced
Spanish	Speaking: Advanced	Writing: Intermediate	Reading: Intermediate